Modelling and control of multibody mechanical systems

Model answers

Question 1

a) The Euler equations of motion for the rigid body with one point fixed are given by

$$\begin{aligned} I_{xx}\dot{\Omega}_x - \Omega_y\Omega_z(I_{yy} - I_{zz}) &= N_x, \\ I_{yy}\dot{\Omega}_y - \Omega_z\Omega_x(I_{zz} - I_{xx}) &= N_y, \\ I_{zz}\dot{\Omega}_z - \Omega_x\Omega_y(I_{xx} - I_{yy}) &= N_z, \end{aligned}$$

[3 marks]

b) Three single-axis-rotation transformation matrices are needed.

$$D_\psi = \begin{bmatrix} \cos\psi & \sin\psi & 0 \\ -\sin\psi & \cos\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

which is the rotation matrix by angle ψ about a z axis.

$$C_\theta = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}$$

which is the rotation matrix by angle θ about a y axis.

Answers

$$B_\phi = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\phi & \sin\phi \\ 0 & -\sin\phi & \cos\phi \end{bmatrix}$$

which is the rotation matrix by angle ϕ about an x axis. The angular velocity vector of the rigid body in the body-fixed coordinate system is

$$\boldsymbol{\Omega} = B_\phi C_\theta D_\psi \begin{bmatrix} 0 \\ 0 \\ \dot{\psi} \end{bmatrix} + B_\phi C_\theta \begin{bmatrix} 0 \\ \dot{\theta} \\ 0 \end{bmatrix} + B_\phi \begin{bmatrix} \dot{\phi} \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} -\dot{\psi}\sin\theta + \dot{\phi} \\ \dot{\psi}\cos\theta\sin\phi + \dot{\theta}\cos\phi \\ \dot{\psi}\cos\theta\cos\phi - \dot{\theta}\sin\phi \end{bmatrix} = \begin{bmatrix} \Omega_x \\ \Omega_y \\ \Omega_z \end{bmatrix}$$

[4 marks]

c) The kinetic energy of the rigid body is

$$T = \frac{1}{2}I_{xx}(-\dot{\psi}\sin\theta + \dot{\phi})^2 + \frac{1}{2}I_{yy}(\dot{\psi}\cos\theta\sin\phi + \dot{\theta}\cos\phi)^2 + \frac{1}{2}I_{zz}(\dot{\psi}\cos\theta\cos\phi - \dot{\theta}\sin\phi)^2$$

As the centre of mass is fixed, the potential energy of the rigid body does not change and therefore the Lagrangian is $L = T$.

[5 marks]

d) The Lagrangian equation of motion corresponding to the generalised coordinate ϕ is

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\phi}}\right) - \frac{\partial L}{\partial \phi} = N_x$$

Evaluating the components of the Lagrangian equation gives

$$\frac{\partial L}{\partial \dot{\phi}} = I_{xx}(-\dot{\psi}\sin\theta + \dot{\phi}) = I_{xx}\Omega_x$$

and

$$\frac{d}{dt}\left(I_{xx}\Omega_x\right) = I_{xx}\dot{\Omega}_x$$

The next term in the Lagrangian equation is

$$\begin{aligned}\frac{\partial L}{\partial \phi} &= I_{yy}(\dot{\psi}\cos\theta\sin\phi + \dot{\theta}\cos\phi)(\dot{\psi}\cos\theta\cos\phi - \dot{\theta}\sin\phi) \\ &\quad + I_{zz}(\dot{\psi}\cos\theta\cos\phi - \dot{\theta}\sin\phi)(-\dot{\psi}\cos\theta\sin\phi - \dot{\theta}\cos\phi)\end{aligned}$$

which results in

$$\frac{\partial L}{\partial \phi} = I_{yy}\Omega_y\Omega_z - I_{zz}\Omega_y\Omega_z$$

Putting everything together yields

$$I_{xx}\dot{\Omega}_x - \Omega_y\Omega_z(I_{yy} - I_{zz}) = N_x$$

which is the first Euler equation shown earlier.

[8 marks]

Answers

Question 2

a) The position vector of the cart is $\boldsymbol{r_M} = x\boldsymbol{i}$ and the position vector of the arm centre of mass is

$$\boldsymbol{r_m} = x\boldsymbol{i} + \frac{l}{2}\boldsymbol{e_r} = \left(x - \frac{l}{2}\sin\theta\right)\boldsymbol{i} - \frac{l}{2}\cos\theta\boldsymbol{k}$$

[3 marks]

b) By differentiating the position vector, the velocity vector of the cart is $\dot{\boldsymbol{r}}_{\boldsymbol{M}} = \dot{x}\boldsymbol{i}$ and of the arm centre of mass is

$$\dot{\boldsymbol{r}}_{\boldsymbol{m}} = \dot{x}\boldsymbol{i} + \frac{l}{2}\dot{\theta}\boldsymbol{e_\theta} = \left(\dot{x} - \frac{l}{2}\dot{\theta}\cos\theta\right)\boldsymbol{i} + \frac{l}{2}\dot{\theta}\sin\theta\boldsymbol{k} = -\dot{x}\sin\theta\boldsymbol{e_r} + \left(-\dot{x}\cos\theta + \frac{l}{2}\dot{\theta}\right)\boldsymbol{e_\theta}$$

[3 marks]

c) The kinetic energy of the system is

$$T = \frac{1}{2}M\dot{\boldsymbol{r}}_{\boldsymbol{M}}\cdot\dot{\boldsymbol{r}}_{\boldsymbol{M}} + \frac{1}{2}m\dot{\boldsymbol{r}}_{\boldsymbol{m}}\cdot\dot{\boldsymbol{r}}_{\boldsymbol{m}} + \frac{1}{2}I\dot{\theta}^2 = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left(\dot{x}^2 - \dot{x}\dot{\theta}l\cos\theta + \frac{l^2}{4}\dot{\theta}^2\right) + \frac{1}{2}I\dot{\theta}^2$$

[3 marks]

d) The horizontal level at O is taken as the zero potential energy level, therefore, the potential energy of the system is

$$V = -m\boldsymbol{r_m}\cdot\boldsymbol{g} = mg\frac{l}{2}\cos\theta$$

Hence the Lagrangian is $L = T - V$, which gives

$$L = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left(\dot{x}^2 - \dot{x}\dot{\theta}l\cos\theta + \frac{l^2}{4}\dot{\theta}^2\right) + \frac{1}{2}I\dot{\theta}^2 - mg\frac{l}{2}\cos\theta$$

[3 marks]

e) The Lagrangian equation with respect to the generalised coordinate x is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial\dot{x}}\right) - \frac{\partial L}{\partial x} = F_c$$

which gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(M\dot{x} + m\dot{x} - m\dot{\theta}\frac{l}{2}\cos\theta\right) = F_c$$

and finally yields

$$(M+m)\ddot{x} - m\frac{l}{2}\ddot{\theta}\cos\theta + m\frac{l}{2}\dot{\theta}^2\sin\theta = F_c$$

The Lagrangian equation with respect to the generalised coordinate θ is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial\dot{\theta}}\right) - \frac{\partial L}{\partial\theta} = T_a$$

which gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(-m\dot{x}\frac{l}{2}\cos\theta + m\frac{l^2}{4}\dot{\theta} + I\dot{\theta}\right) - m\dot{x}\dot{\theta}\frac{l}{2}\sin\theta - mg\frac{l}{2}\sin\theta = T_a$$

Answers

and finally gives

$$-m\frac{l}{2}\ddot{x}\cos\theta + \left(m\frac{l^2}{4} + I\right)\ddot{\theta} - mg\frac{l}{2}\sin\theta = T_a$$

[8 marks]

Answers

Question 3

a) The generalised coordinates are θ and x. [2 marks]

b) The velocity vector of the cart is $\dot{\boldsymbol{r}}_{\boldsymbol{M}} = \dot{x}\boldsymbol{i}$. By differentiating the velocity expression, the acceleration vector of the cart is given by

$$\ddot{\boldsymbol{r}}_{\boldsymbol{M}} = \ddot{x}\boldsymbol{i}$$

The velocity vector of the arm centre of mass is $\dot{\boldsymbol{r}}_{\boldsymbol{m}} = \dot{x}\boldsymbol{i} + \frac{l}{2}\dot{\theta}\boldsymbol{e}_{\boldsymbol{\theta}} = \left(\dot{x} - \frac{l}{2}\dot{\theta}\cos\theta\right)\boldsymbol{i} + \frac{l}{2}\dot{\theta}\sin\theta\boldsymbol{k} = -\dot{x}\sin\theta\boldsymbol{e}_{\boldsymbol{r}} + \left(-\dot{x}\cos\theta + \frac{l}{2}\dot{\theta}\right)\boldsymbol{e}_{\boldsymbol{\theta}}$. By differentiating the velocity expression, the acceleration vector of the arm centre of mass is given by

$$\ddot{\boldsymbol{r}}_{\boldsymbol{m}} = \ddot{x}\boldsymbol{i} + \frac{l}{2}\ddot{\theta}\boldsymbol{e}_{\boldsymbol{\theta}} - \frac{l}{2}\dot{\theta}^2\boldsymbol{e}_{\boldsymbol{r}} = \left(\ddot{x} - \frac{l}{2}\ddot{\theta}\cos\theta + \frac{l}{2}\dot{\theta}^2\sin\theta\right)\boldsymbol{i} + \left(\frac{l}{2}\ddot{\theta}\sin\theta + \frac{l}{2}\dot{\theta}^2\cos\theta\right)\boldsymbol{k}$$

$$= \left(-\ddot{x}\sin\theta - \frac{l}{2}\dot{\theta}^2\right)\boldsymbol{e}_{\boldsymbol{r}} + \left(-\ddot{x}\cos\theta + \frac{l}{2}\ddot{\theta}\right)\boldsymbol{e}_{\boldsymbol{\theta}}.$$

[3 marks]

c) The total force vector on the cart is

$$\boldsymbol{F}_{\text{cart}} = F_c\boldsymbol{i} - F_x\boldsymbol{i} - F_z\boldsymbol{k} + Mg\boldsymbol{k} - R\boldsymbol{k}$$

where R is the normal reaction from the horizontal surface on the cart. The total force vector on the arm is

$$\boldsymbol{F}_{\text{arm}} = F_x\boldsymbol{i} + F_z\boldsymbol{k} + mg\boldsymbol{k}$$

Answers

[3 marks]

d) By considering the translational motion of the centre of mass of the arm one has

$$F_x\boldsymbol{i} + F_z\boldsymbol{k} + mg\boldsymbol{k} = m\left(\left(\ddot{x} - \frac{l}{2}\ddot{\theta}\cos\theta + \frac{l}{2}\dot{\theta}^2\sin\theta\right)\boldsymbol{i} + \left(\frac{l}{2}\ddot{\theta}\sin\theta + \frac{l}{2}\dot{\theta}^2\cos\theta\right)\boldsymbol{k}\right)$$

Therefore, by collecting the $\boldsymbol{i}$ terms

$$F_x = m\left(\ddot{x} - \frac{l}{2}\ddot{\theta}\cos\theta + \frac{l}{2}\dot{\theta}^2\sin\theta\right) \tag{1}$$

and by collecting the $\boldsymbol{k}$ terms

$$F_z = m\left(\frac{l}{2}\ddot{\theta}\sin\theta + \frac{l}{2}\dot{\theta}^2\cos\theta - g\right) \tag{2}$$

[5 marks]

e) By considering the translational motion of the cart one has

$$F_c\boldsymbol{i} - F_x\boldsymbol{i} - F_z\boldsymbol{k} + Mg\boldsymbol{k} - R\boldsymbol{k} = M\ddot{x}\boldsymbol{i}$$

Therefore, by collecting the $\boldsymbol{i}$ terms and also substituting F_x from (1), gives the first equation of motion as

$$(M+m)\ddot{x} - m\frac{l}{2}\ddot{\theta}\cos\theta + m\frac{l}{2}\dot{\theta}^2\sin\theta = F_c$$

By considering the motion about the centre of mass of the arm according to $\frac{d\boldsymbol{H}}{dt} = \boldsymbol{N}$, where

$$H = I\dot{\theta}$$

and

$$N = \frac{l}{2}\cos\theta F_x - \frac{l}{2}\sin\theta F_z + T_a$$

gives

$$I\ddot{\theta} = \frac{l}{2}\cos\theta F_x - \frac{l}{2}\sin\theta F_z + T_a$$

By substitution, using equations (1) and (2), the second equation of motion is

$$-m\frac{l}{2}\ddot{x}\cos\theta + \left(m\frac{l^2}{4} + I\right)\ddot{\theta} - mg\frac{l}{2}\sin\theta = T_a$$

[7 marks]

Answers

Question 4

a) i)

$$\boldsymbol{r_i} = \boldsymbol{R} + \boldsymbol{r'_i} \tag{3}$$

in which $\boldsymbol{R}$ is the position vector of the centre of mass, C, of the rigid body with respect to O and $\boldsymbol{r'_i}$ is the position vector of the ith mass particle of the rigid body with respect to the rigid body centre of mass, C.

ii)

$$\boldsymbol{R} = \boldsymbol{R_1} + \boldsymbol{R_2} \tag{4}$$

in which $\boldsymbol{R_1}$ is the position vector of point A of the rigid body with respect to O and $\boldsymbol{R_2}$ is the position vector of the centre of mass, C, of the rigid body with respect to point A.

iii)

$$\boldsymbol{r_i} = \boldsymbol{R_1} + \boldsymbol{R_2} + \boldsymbol{r'_i} \tag{5}$$

[3 marks]

b) i) By differentiating (3)

$$\frac{d\boldsymbol{r_i}}{dt} = \boldsymbol{v_i} = \dot{\boldsymbol{R}} + \boldsymbol{v_i^R} = \dot{\boldsymbol{R}} + \boldsymbol{\Omega} \times \boldsymbol{r'_i}$$

in which $\dot{\boldsymbol{R}}$ is the velocity vector of the rigid body centre of mass relative to the fixed point O, and

$$\boldsymbol{v_i^R} = \frac{d\boldsymbol{r'_i}}{dt}$$

is the velocity vector of the ith mass particle relative to the centre of mass of the rigid body.

Answers

ii) By differentiating (5)

$$\frac{d\boldsymbol{r_i}}{dt} = \boldsymbol{v_i} = \dot{\boldsymbol{R}}_{\boldsymbol{1}} + \dot{\boldsymbol{R}}_{\boldsymbol{2}} + \boldsymbol{v_i^R} = \dot{\boldsymbol{R}}_{\boldsymbol{1}} + \dot{\boldsymbol{R}}_{\boldsymbol{2}} + \boldsymbol{\Omega} \times \boldsymbol{r'_i} \tag{6}$$

in which $\dot{\boldsymbol{R}}_{\boldsymbol{1}}$ is the velocity vector of point A relative to the fixed point O and $\dot{\boldsymbol{R}}_{\boldsymbol{2}}$ is the velocity vector of the centre of mass, C, relative to point A.

[4 marks]

c) By using (6) in the kinetic energy expression

$$T = \frac{1}{2}\sum_i m_i v_i^2 = \frac{1}{2}\sum_i m_i \boldsymbol{v_i} \cdot \boldsymbol{v_i} = \frac{1}{2}\sum_i m_i \left(\dot{\boldsymbol{R}}_{\boldsymbol{1}} + \dot{\boldsymbol{R}}_{\boldsymbol{2}} + \boldsymbol{v_i^R}\right) \cdot \left(\dot{\boldsymbol{R}}_{\boldsymbol{1}} + \dot{\boldsymbol{R}}_{\boldsymbol{2}} + \boldsymbol{v_i^R}\right)$$

and expanding

$$T = \frac{1}{2}\sum_i m_i |\dot{\boldsymbol{R}}_{\boldsymbol{1}}|^2 + \frac{1}{2}\sum_i m_i |\dot{\boldsymbol{R}}_{\boldsymbol{2}}|^2 + \sum_i m_i \dot{\boldsymbol{R}}_{\boldsymbol{1}} \cdot \dot{\boldsymbol{R}}_{\boldsymbol{2}} + \frac{1}{2}\sum_i m_i v_i^{R^2} + \left(\dot{\boldsymbol{R}}_{\boldsymbol{1}} + \dot{\boldsymbol{R}}_{\boldsymbol{2}}\right) \cdot \frac{d}{dt}\left(\sum_i m_i \boldsymbol{r'_i}\right)$$

The last term in the right-hand side contains the factor $\sum m_i \boldsymbol{r'_i}$ which is a null vector because it defines the position vector of the centre of mass in the coordinate system

whose origin is the centre of mass. Therefore the last term vanishes and after further simplification

$$T = \frac{1}{2}M\left(|\dot{\boldsymbol{R}}_{\boldsymbol{1}}|^2 + |\dot{\boldsymbol{R}}_{\boldsymbol{2}}|^2 + 2\dot{\boldsymbol{R}}_{\boldsymbol{1}} \cdot \dot{\boldsymbol{R}}_{\boldsymbol{2}}\right) + \frac{1}{2}\sum_i m_i v_i^{R^2}$$

in which the last term is the kinetic energy of the motion about the centre of mass and, as in the notes, it is given by

$$\frac{1}{2}\sum_i m_i v_i^{R^2} = \frac{\boldsymbol{\Omega}}{2} \cdot \sum_i m_i(\boldsymbol{r}_{\boldsymbol{i}}' \times \boldsymbol{v}_{\boldsymbol{i}}^{\boldsymbol{R}}) = \frac{\boldsymbol{\Omega} \cdot \boldsymbol{H}_{\boldsymbol{0}}}{2}$$

with $\boldsymbol{H}_{\boldsymbol{0}}$ the angular momentum of motion about the centre of mass.

[5 marks]

d) Starting from $\frac{d\boldsymbol{H}}{dt} = \boldsymbol{R} \times M\frac{d^2\boldsymbol{R}}{dt^2} + \frac{d\boldsymbol{H}_{\boldsymbol{0}}}{dt}$ given in the text of the question, and substituting (4) gives

$$\frac{d\boldsymbol{H}}{dt} = (\boldsymbol{R}_{\boldsymbol{1}} + \boldsymbol{R}_{\boldsymbol{2}}) \times M\frac{d^2\left(\boldsymbol{R}_{\boldsymbol{1}} + \boldsymbol{R}_{\boldsymbol{2}}\right)}{dt^2} + \frac{d\boldsymbol{H}_{\boldsymbol{0}}}{dt} \tag{7}$$

Then substituting (5) into $\boldsymbol{N}^{(e)} = \sum_i \boldsymbol{r}_{\boldsymbol{i}} \times \boldsymbol{F}_{\boldsymbol{i}}^{(e)}$, which is also given in the text of the question, yields

$$\boldsymbol{N}^{(e)} = \sum_i \left(\boldsymbol{R}_{\boldsymbol{1}} + \boldsymbol{R}_{\boldsymbol{2}} + \boldsymbol{r}_{\boldsymbol{i}}'\right) \times \boldsymbol{F}_{\boldsymbol{i}}^{(e)} = \boldsymbol{R}_{\boldsymbol{1}} \times \boldsymbol{F}^{(e)} + \sum_i \left(\boldsymbol{R}_{\boldsymbol{2}} + \boldsymbol{r}_{\boldsymbol{i}}'\right) \times \boldsymbol{F}_{\boldsymbol{i}}^{(e)}$$

where the last term in the right-hand side is the total moment about point A of external forces acting on the rigid body

$$\boldsymbol{N}_A^{(e)} = \sum_i \left(\boldsymbol{R}_{\boldsymbol{2}} + \boldsymbol{r}_{\boldsymbol{i}}'\right) \times \boldsymbol{F}_{\boldsymbol{i}}^{(e)}$$

Answers

Therefore, by also considering $M\frac{d^2\boldsymbol{R}}{dt^2} = \boldsymbol{F}^{(e)}$, which is given in the text of the question, and also using (4)

$$\boldsymbol{N}^{(e)} = \boldsymbol{R}_{\boldsymbol{1}} \times M\frac{d^2\left(\boldsymbol{R}_{\boldsymbol{1}} + \boldsymbol{R}_{\boldsymbol{2}}\right)}{dt^2} + \boldsymbol{N}_A^{(e)} \tag{8}$$

Substituting (7) and (8) into $\frac{d\boldsymbol{H}}{dt} = \boldsymbol{N}^{(e)}$ (equation (4.2) in the text of the question), and simplifying, yields the equation for the rotational motion of the rigid body about point A

$$\boldsymbol{R}_{\boldsymbol{2}} \times M\frac{d^2\left(\boldsymbol{R}_{\boldsymbol{1}} + \boldsymbol{R}_{\boldsymbol{2}}\right)}{dt^2} + \frac{d\boldsymbol{H}_{\boldsymbol{0}}}{dt} = \boldsymbol{N}_A^{(e)}$$

which can be rewritten as

$$\boldsymbol{R}_{\boldsymbol{2}} \times M\frac{d^2\boldsymbol{R}}{dt^2} + \frac{d\boldsymbol{H}_{\boldsymbol{0}}}{dt} = \boldsymbol{N}_A^{(e)}$$

in which the first term on the left-hand side corresponds to the moment produced about point A by the inertial force of the rigid body centre of mass.

[8 marks]